

Research Interests

Human-AI Interaction • Computational Social Science • Digital Mental Health and Well-being • Social Computing • Natural Language Processing • Human-Computer Interaction

Education

University of Wisconsin | Madison, WI May 2024
M.S. Computer Sciences GPA: 3.66/4.00

- Coursework: Human-Computer Interaction, Data Visualization, Modeling User Interaction, Big Data Systems, Algorithms, Communication Theory & Research, Teaching STEM

University of Michigan | Ann Arbor, MI Dec 2021
B.S.E. Computer Science; B.S. Cognitive Science; Minor in Complex Systems GPA: 3.54/4.00

- Coursework: Machine Learning, Artificial Intelligence, Learning & Memory, Computational Modeling of Cognition, Software Engineering, Network Science

Research Experience

Research Volunteer Jul 2026 – present
CHAT Lab, Singapore Management University

- Designed, built, and deployed a [global social media policy map](#) (JavaScript, D3.js, Vercel) tracking 130+ government policies on minors' access across 100+ countries for an upcoming preprint, with live updates from the team's dataset, a rollout timeline, multi-attribute filtering, and side-by-side comparison.
- Developing companion dashboards tracking [AI mental-health policy](#) and [platforms' own youth-safety policies](#), extending the same live-updating framework to new datasets.
- Conducting a systematic umbrella review synthesizing evidence on mental health chatbot interventions.

Data Science Research Fellow Jan 2025 – present
U.S. Department of Veterans Affairs (VA)

- Designed and built end-to-end Python and SQL pipelines that turn 1.25M+ unstructured clinical records into structured, analysis-ready research datasets, with automated validation and post-processing.
- Collaborated with clinicians and interdisciplinary researchers to define research variables and validate computational measures; designed evaluation protocols assessing their reliability and validity.
- Deployed open-weight large language models of two sizes across an 8-GPU cluster using vLLM with 4-bit quantization and pipeline parallelism, making large-model deployment feasible on fixed hardware.
- Fine-tuned a transformer classifier as an upstream filter that reduced downstream hallucination, built evaluation tooling that categorizes hallucination, omission, and mismatch errors against 5,486 labeled examples, and benchmarked model sizes on accuracy and cost, presenting results to leadership.
- Maintained a SQL database drawing on an enterprise data warehouse for program reporting; built R dashboards for VA Office of Rural Health telehealth evaluation and Streamlit dashboards for stakeholders.

COVID Twitter Research Oct 2022 – Dec 2023
University of Wisconsin

- Analyzed 10M+ COVID-related tweets using structural topic modeling in R, producing visualizations and reports showing shifts in public discourse over time during the pandemic.

Parsons Problem Research Oct 2020 – Aug 2021
University of Michigan, co-mentored by Prof. Xu Wang and Prof. Barbara Ericson

- Helped develop and test a tool that recommended personalized exercises based on students' in-progress work.
- Designed and conducted think-aloud studies with students to understand their problem-solving process.
- Extracted and analyzed data from CSV logs to discover patterns and measure performance of student code.

Teaching

Graduate Teaching Assistant

Aug 2022 – May 2024

University of Wisconsin, Computer Sciences

- Supported hundreds of students in “Building User Interfaces” and “Human-Computer Interaction” through office hours, forum support, course design, exam and homework development, grading, and feedback.

Additional Experience

Natural Language Modeling Intern

May 2022 – Aug 2022

Licens.io

- Built sentiment classifiers for GitHub issue comments using spaCy transformers, reaching state-of-the-art accuracy, and curated a labeled dataset of comments from a random sample of R projects.

Cloud Software Engineering Intern

Jun 2021 – Aug 2021

American Century Investments

- Engineered AWS Lambda functions for data replication monitoring and automated alerting.

Leadership

Student Chapter of ACM (SACM) at UW-Madison

Sep 2023 – May 2024

Officer

- Planned and organized CS graduate community and department events, including prospective student visits.

Wisconsin Union Directorate - Distinguished Lecture Series

Sep 2022 – May 2024

Associate Director of Marketing and Social Media

- Wrote a weekly newsletter for over 2,000 subscribers, managed social media, designed graphics, and coordinated outreach for lectures drawing audiences of over 1,000.

Cognitive Science Community at University of Michigan

Sep 2020 – Dec 2021

Vice President

- Revamped the website and graphics, managed the leadership team, and led colloquium planning.

Skills

Research Methods	Survey design, human-subjects research, think-aloud studies, qualitative coding, evidence synthesis, experimental design, systematic review methodology
Programming	Python (pandas, scikit-learn, PyTorch, spaCy, transformers), R (tidyverse, stm, ggplot2), SQL, JavaScript (D3.js, TopoJSON), Docker
Computational	Natural language processing, machine learning, computational text analysis, structural topic modeling, statistical analysis, information extraction
LLM Tools	vLLM, AWQ quantization, LangChain, LlamaIndex, n8n, AI-assisted coding tools
Software	Qualtrics, Git, L ^A T _E X, Jupyter, Tableau, Streamlit

Publications & Manuscripts

- Weerasinghe, N.**, Razjouyan, J., Sharafkhaneh, A., & Maghsoudi, A. (2026). Automated extraction of narcolepsy and hypersomnia diagnostic metrics from clinical notes using large language models. *Manuscript in preparation*.

Conference Presentations

- Weerasinghe, N.**, Sharafkhaneh, S., Sharafkhaneh, S., Kaleekal, C., Bandi, A., Chandler, B., Razjouyan, J., & Maghsoudi, A. (2026). Comparing bag of words and transformer models to identify MSLT reports in VA clinical notes. Late-Breaking Abstract, SLEEP 2026.
- Weerasinghe, N.**, Balachandran, S., Aminnia, S., Razjouyan, J., & Sharafkhaneh, A. (2026). Diverging effects of hypoxic burden on cardiovascular risk and mortality in patients with obstructive sleep apnea syndrome. ATS 2026.
- Maghsoudi, A., **Weerasinghe, N.**, Sharafkhaneh, A., & Razjouyan, J. (2025). Mean sleep latency extraction using machine learning and LLM pipelines. IEEE AIXSET 2025.